

Compare and order large numbers

Use the scaffold, show each step and ask for an adult check when marked.

Name: _____ Date: _____

1. Read the model: Compare 5,906,120 and 5,960,102. Copy or complete the first step.

2. Order 7,405,090, 7,450,009 and 7,045,900 from least to greatest.

3. Insert $<$, $>$ or $=$: 9,099,999 ___ 9,100,001.

4. Miro says: Miro compares the final digits first because they are on the right. Is this correct? Explain.

Teacher answers and checking prompts

1. Read the model: Compare 5,906,120 and 5,960,102. Copy or complete the first step.
Answer 5,906,120 < 5,960,102.
2. Order 7,405,090, 7,450,009 and 7,045,900 from least to greatest.
Answer 7,045,900, 7,405,090, 7,450,009.
3. Insert <, > or =: 9,099,999 ___ 9,100,001.
Answer 9,099,999 < 9,100,001.
4. Miro says: Miro compares the final digits first because they are on the right. Is this correct? Explain.
Answer Compare from the greatest place value on the left and stop at the first different digit.

Compare and order large numbers

Work independently. Show a complete method and check at least one answer.

Name: _____ Date: _____

1. Compare 5,906,120 and 5,960,102.

6. Check this result using an inverse, estimate or known fact:
Insert $<$, $>$ or $=$: 9,099,999 ___ 9,100,001.

2. Order 7,405,090, 7,450,009 and 7,045,900 from least to greatest.

7. Find a whole number that lies between 6,799,999 and 6,800,010.

3. Insert $<$, $>$ or $=$: 9,099,999 ___ 9,100,001.

8. Identify and correct this misconception: Miro compares the final digits first because they are on the right.

4. Solve this independently and show every stage: Compare 5,906,120 and 5,960,102.

9. Write a complete sentence explaining how you checked one answer.

5. Use a second method or representation for: Order 7,405,090, 7,450,009 and 7,045,900 from least to greatest.

10. Write and solve a similar question to: Insert $<$, $>$ or $=$: 9,099,999 ___ 9,100,001.

Teacher answers and checking prompts

1. Compare 5,906,120 and 5,960,102.
Answer 5,906,120 < 5,960,102.
2. Order 7,405,090, 7,450,009 and 7,045,900 from least to greatest.
Answer 7,045,900, 7,405,090, 7,450,009.
3. Insert <, > or =: 9,099,999 ___ 9,100,001.
Answer 9,099,999 < 9,100,001.
4. Solve this independently and show every stage: Compare 5,906,120 and 5,960,102.
Answer 5,906,120 < 5,960,102.
5. Use a second method or representation for: Order 7,405,090, 7,450,009 and 7,045,900 from least to greatest.
Answer Expected result: 7,045,900, 7,405,090, 7,450,009. Method or representation may vary.
6. Check this result using an inverse, estimate or known fact: Insert <, > or =: 9,099,999 ___ 9,100,001.
Answer Expected result: 9,099,999 < 9,100,001. A valid check must be shown.
7. Find a whole number that lies between 6,799,999 and 6,800,010.
Answer Any integer from 6,800,000 to 6,800,009, for example 6,800,004.
8. Identify and correct this misconception: Miro compares the final digits first because they are on the right.
Answer Compare from the greatest place value on the left and stop at the first different digit.
9. Write a complete sentence explaining how you checked one answer.
Answer Answer should name an inverse, estimate, boundary, known fact or equivalent representation.
10. Write and solve a similar question to: Insert <, > or =: 9,099,999 ___ 9,100,001.
Answer Answers vary. The new question must use the same mathematical structure and include a correct solution.

Compare and order large numbers

Prove, compare or generalise. Write enough reasoning for another pupil to follow.

Name: _____ Date: _____

1. Find a whole number that lies between 6,799,999 and 6,800,010.

6. Create a new example that tests the same idea as: Insert $<$, $>$ or $=$: 9,099,999 ___ 9,100,001.

2. Prove the answer to this question, rather than only calculating it: Compare 5,906,120 and 5,960,102.

7. Find a second method or representation. Compare its efficiency with your first method.

3. Solve this in two different ways and compare them: Order 7,405,090, 7,450,009 and 7,045,900 from least to greatest.

8. Write one always, sometimes or never statement about today's learning and justify it.

4. Work backwards from the answer to reconstruct the question: $9,099,999 < 9,100,001$.

9. Create a believable incorrect solution. Identify the exact step where the reasoning breaks.

5. Explain why this reasoning fails: Miro compares the final digits first because they are on the right.

10. Change one value or condition in a question. Explain what changes in the solution and what stays the same.

Teacher answers and checking prompts

1. Find a whole number that lies between 6,799,999 and 6,800,010.
Answer Any integer from 6,800,000 to 6,800,009, for example 6,800,004.
2. Prove the answer to this question, rather than only calculating it: Compare 5,906,120 and 5,960,102.
Answer Expected result: $5,906,120 < 5,960,102$. Proof or justification must match the lesson structure.
3. Solve this in two different ways and compare them: Order 7,405,090, 7,450,009 and 7,045,900 from least to greatest.
Answer Expected result: 7,045,900, 7,405,090, 7,450,009. Two valid methods and a comparison are required.
4. Work backwards from the answer to reconstruct the question: $9,099,999 < 9,100,001$.
Answer One valid reconstruction is: Insert $<$, $>$ or $=$: $9,099,999 ___ 9,100,001$. Other equivalent questions are acceptable.
5. Explain why this reasoning fails: Miro compares the final digits first because they are on the right.
Answer Compare from the greatest place value on the left and stop at the first different digit.
6. Create a new example that tests the same idea as: Insert $<$, $>$ or $=$: $9,099,999 ___ 9,100,001$.
Answer Answers vary. The example and solution must preserve the mathematical structure.
7. Find a second method or representation. Compare its efficiency with your first method.
Answer Answers vary. Comparison should refer to accuracy, number structure and clarity.
8. Write one always, sometimes or never statement about today's learning and justify it.
Answer Answers vary. A valid example and counterexample should support the classification.
9. Create a believable incorrect solution. Identify the exact step where the reasoning breaks.
Answer Answers vary. The error must be mathematically relevant and the correction must be precise.
10. Change one value or condition in a question. Explain what changes in the solution and what stays the same.
Answer Answers vary. The explanation must distinguish the mathematical structure from the changed value.